# Alex Hsiao (Wei-Chun Hsiao)

**FULL-STACK GENAI ENGINEER**

Currently serving as a **Full-Stack GenAI Engineer** at Tatung System Technologies, with hands-on expertise spanning full-stack application development, LLM agents and RAG pipelines, containerized GPU infrastructure (NVIDIA GPU Operator & HAMi), stateful Multi-Agent LLMs with LangGraph, distributed microservices, and ISO/IEC 27001 security compliance.

**1M** tokens usable context on one H100 **0** vulnerabilities · 6 hardened images **95.3%** AI Agent accuracy · 300 questions

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STATUS
ONLINE // OPEN FOR SENIOR ROLES

## Experience

**NOV 2025 – PRESENT**

**Tatung System Technologies Inc. (Tatung World Tech)**

Full-Stack GenAI Engineer

Zhongshan Dist., Taipei

- Built an enterprise property-leasing management system for a real-estate leasing client — 13 functional modules and 21 sidebar pages — owning system design, application development, and infrastructure end to end: hybrid AI Agent, document search, tool calling, **RAG** with hybrid vector + BM25 (PGroonga) retrieval across 8 business entity types, S3-compatible file management, GIS coordinates, agent-based SQL querying, and automated Agent evaluation. Enforced data access through dual-layer guardrails (Row-Level Security + read-only DB account).
- Built an internal material automation system that reads purchase orders and quotations and converts them into SAP material master data, replacing manual entry for 113 users: median extraction time 41 seconds versus 15 minutes by hand (~22× faster; 4 of 22 batches routed to manual review).
- Extended a 120B model's usable context from ~30K to 1M tokens on a single H100 by offloading the KV cache to CPU RAM (140 GB) with FP8 KV-cache quantization and Multi-Token Prediction speculative decoding — verified at 990,016 tokens in one request, 10/10 runs successful with zero preemptions.
- Deployed local LLMs on a client's **RTX PRO 6000** GPUs with Multi-Token Prediction, serving inference via vLLM and llama.cpp in parallel on the same Kubernetes cluster across multiple GPUs.

**STACK** · React.js, FastAPI, LangGraph, vLLM, llama.cpp, Azure OpenAI, PGroonga, Valkey, RabbitMQ, PostgreSQL, Kubernetes, Kubespray, Helm, NVIDIA GPU Operator, HAMi, Longhorn, CrowdSec, OpenBao

**SEP 2026 – DEC 2026 (IN PROGRESS)**

**Gtalent Education (大數智能教育)**

Technical Instructor — NVIDIA NCA-GENL

Live online

- Teaching the NVIDIA NCA-GENL generative AI associate certification program: 5 modules and 32 one-hour live-online sessions delivered on weekends (15 of 32 sessions completed as of Nov 2026).

**SEP 2025 – JUN 2026**

**AIGC Research Club, National Chiayi University**

Faculty Advisor / Technical Consultant

Chiayi City

- Invited as faculty advisor and guest lecturer during the club's founding period, helping define its AIGC research direction and academic ethics guidelines.

**MAR 2025 – NOV 2025 (9 MOS)**

**National Institute of Cyber Security (NICS)**

Software Engineer

Zhongzheng Dist., Taipei

- Drove cloud security initiatives and contributed directly to the development of the **AWS Government Configuration Baseline** for national infrastructure.
- Researched and implemented cutting-edge AI features and autonomous workflows using LangGraph and Multi-Agent LLMs.

**STACK** · Python, Next.js, LangGraph, FastAPI, CloudFormation, MongoDB, AWS Security Baselines

JUN 2024 – MAR 2025 (10 MOS)

**Cloud Service Provider (Agency)**

Project Engineer / Software Engineer

Nangang Dist., Taipei

- Architected and implemented enterprise GCP and AWS cloud environments, executed complex cross-cloud migrations, and configured secure site-to-site VPNs.
- Designed centralized monitoring platforms and deployed production Kubernetes clusters with fine-grained RBAC security policies.

**STACK** · Kubernetes, GCP, AWS, Terraform, Ansible, WAF, Cloud Architecture

SEP 2021 – OCT 2023 (2 YRS 2 MOS)

**Information Security Lab, NCYU CSIE**

Master's Researcher & Project Assistant

Chiayi City, Taiwan

- Conducted advanced research in deep learning and cybersecurity intrusion detection for web application security.
- Designed a novel hybrid architecture combining Multi-Model CNN, Bidirectional GRU, and Multi-Head Self-Attention for SQLi and XSS attack detection.

**STACK** · Python, PyTorch, TensorFlow, Django, Flask, Deep Learning, IEEE Publication

## Technical Skills

**GenAI & Machine Learning**

vLLM Inference Engine · LangGraph Multi-Agents · RAG & Hybrid Retrieval · LLM Agent Evaluation · HAMi vGPU Slicing · NVIDIA GPU Operator · PyTorch & TensorFlow

**FullStack & Distributed Systems**

FastAPI / Python · Node.js (Async Runtimes) · RabbitMQ (AMQP Broker) · Django / Flask · PHP / Laravel

**Cloud Fabric & Orchestration**

Kubernetes (CKA Certified) · AWS Cloud (SAP Certified) · Google Cloud (PCA Certified) · Terraform (IaC Automation) · Docker & Podman

**Databases & Resilient Storage**

PostgreSQL · Microsoft SQL Server · MongoDB & MySQL · Longhorn HA Block Storage · MinIO / S3 Parquet

## Certifications & Publications



**NVIDIA Certified Professional: Generative AI LLMs**

NVIDIA // CREDLY VERIFIED · verify



**Certified Kubernetes Administrator (CKA)**

LINUX FOUNDATION / CNCF · verify



**AWS Certified Solutions Architect – Professional (SAP)**

AMAZON WEB SERVICES · verify



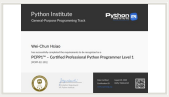
**Google Cloud Professional Cloud Architect (PCA)**

GOOGLE CLOUD · verify



**ISO/IEC 27001 Lead Auditor (Information Security)**

TKSG // ISMS GOVERNANCE · verify



**PCPP1™ – Certified Professional Python Programmer Level 1**

OPENEDG PYTHON INSTITUTE // CREDLY VERIFIED · verify



**IEEE Xplore: Deep Learning SQLi & XSS Detection**

IEEE XPLORE // PEER REVIEWED · verify

## Education

**M.S. in Computer Science & Information Engineering — National Chiayi University**

Master's thesis published in IEEE Xplore (Doc. 10210155): a hybrid CNN / BiGRU / multi-head attention architecture for SQLi and XSS detection.